

Lecture 10
Text Mining and Singular Value Decomposition

October 20, 2026

Data Mining and Time Series Forecasting

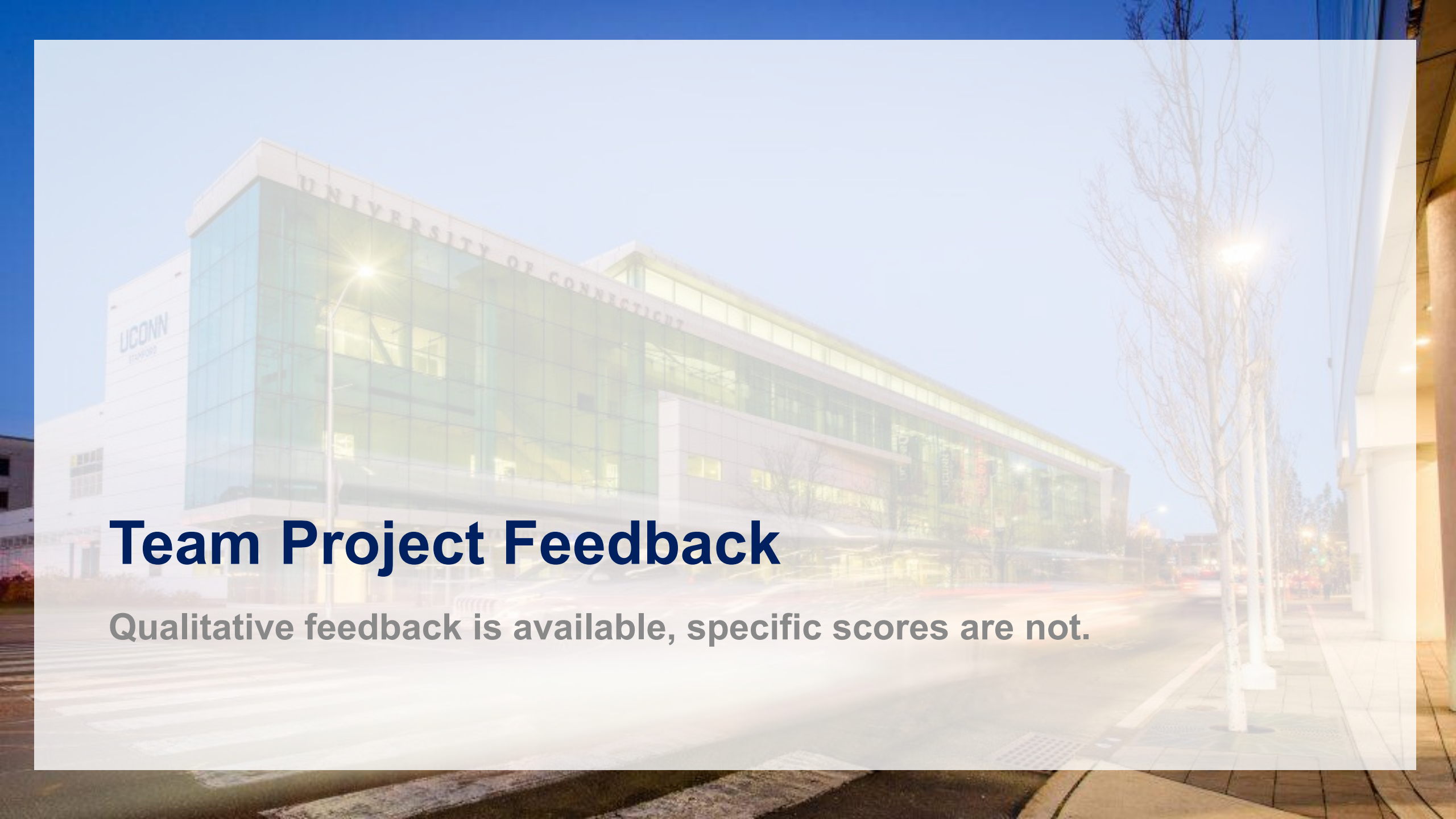
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Agenda

- **Team Project Feedback**
- **Motivation**
- **From Text to Tokens**
  - Text Parsing, Tokenization, and Preprocessing
- **From Text to Numbers**
  - Bag-of-words and Document-term Matrices
- **Weighting Words**
  - Word Frequencies, Zipf's Law, and TF-IDF
- **Singular Value Decomposition**
  - SVD and Lower-dimensional Document Representations
- **Announcements**



Team Project Feedback

Qualitative feedback is available, specific scores are not.



Motivation

Why text mining still maters in the age of LLMs

What's the Point of Learning Text Mining?

- **"Can't we just ask ChatGPT/Claude?"**
  - **LLMs are powerful in**
    - Summarization, contextual interpretation, and question answering.
  - **Business analytics also requires**
    - Measuring patterns, exact matching, comparisons, and reproducible numerical evidence.

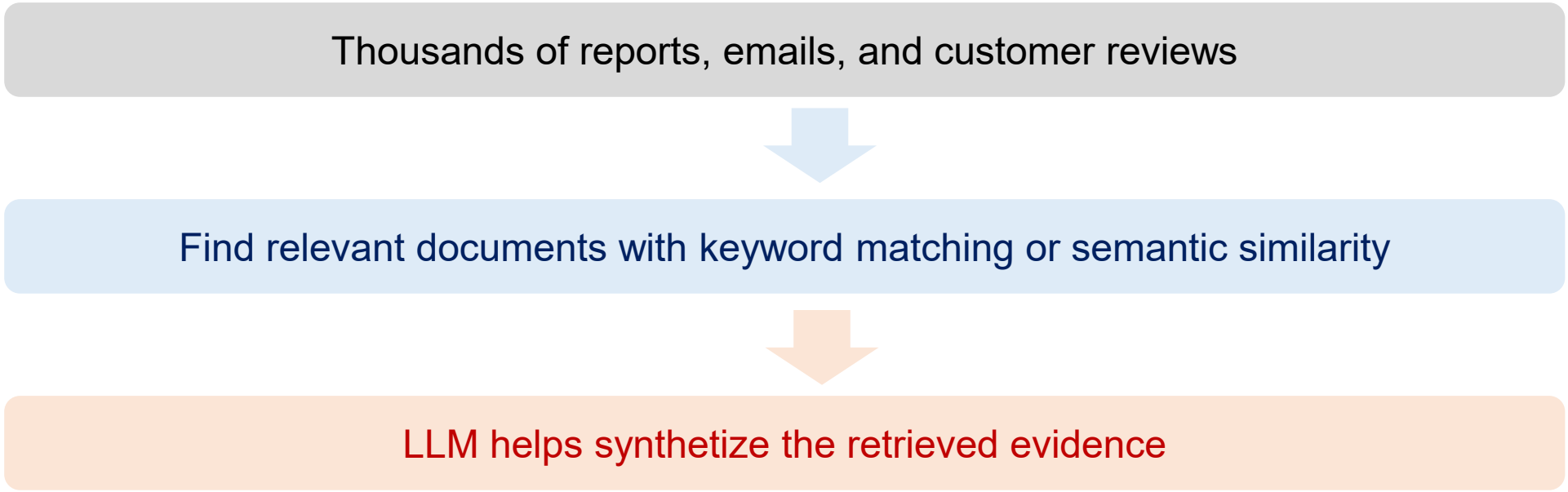
- **More realistic question:**
  - ***"Which representation and tool best serve the business objective?"***

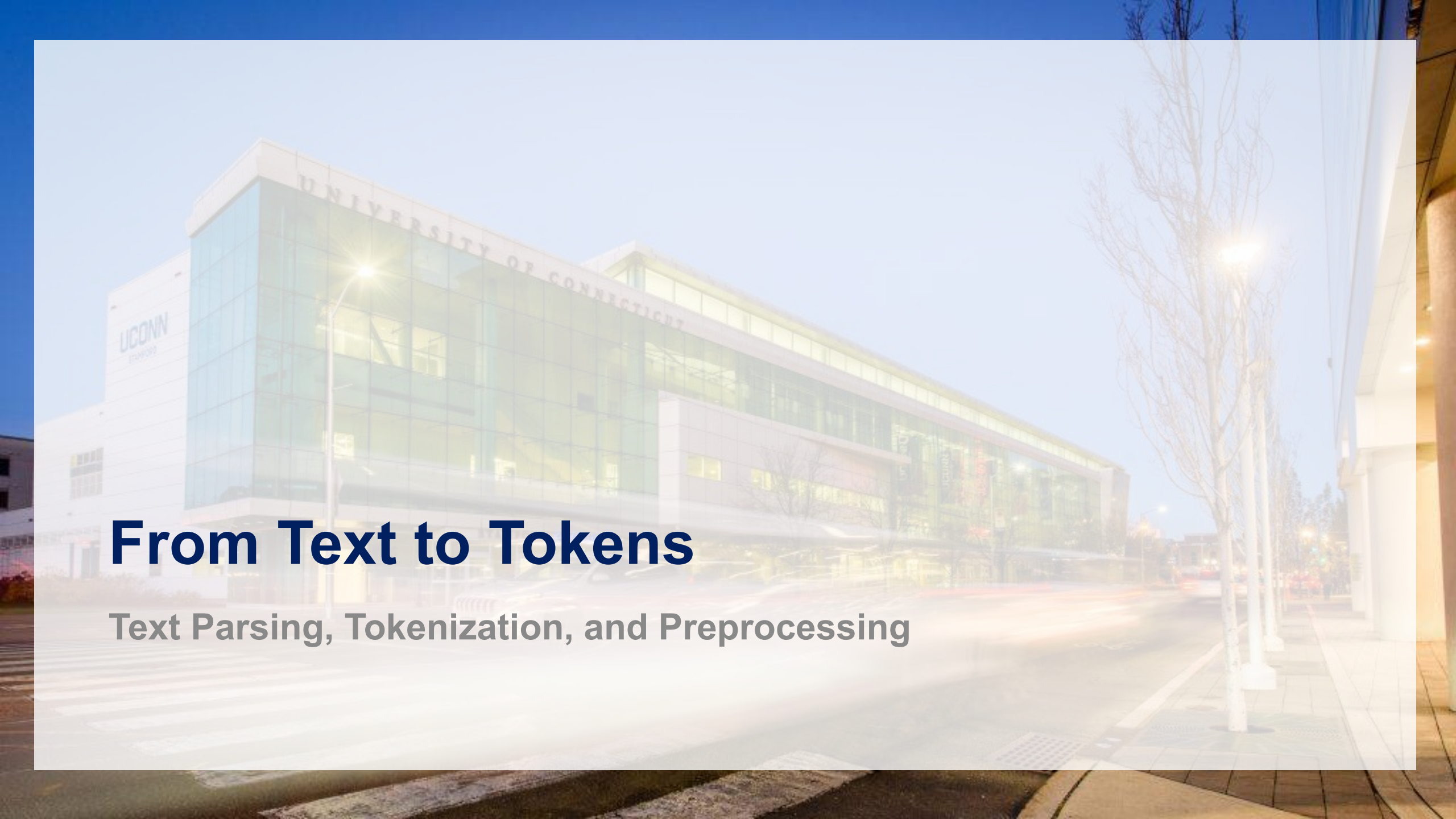
What's the Point of Learning Text Mining?

- **Why are traditional methods still useful?**
  - **Transparent measures**
    - Counts, terms, and weights have explicit definitions. We can audit exactly what was counted.
  - **Practical efficiency**
    - For many tasks, sparse numerical representations are inexpensive to store, search, and analyze.
  - **Strong benchmarks**
    - A simple baseline provides evidence that a more complex solution actually adds value.

What's the Point of Learning Text Mining?

- Classical methods and LLMs can work together.
- Example:
  - Elastic Docs, Hybrid search and RAG (Search with Elasticsearch | Elastic Docs)





From Text to Tokens

Text Parsing, Tokenization, and Preprocessing

A Running Example: Customer Support Tickets

- A company wants to identify recurring customer problems without reading every ticket.

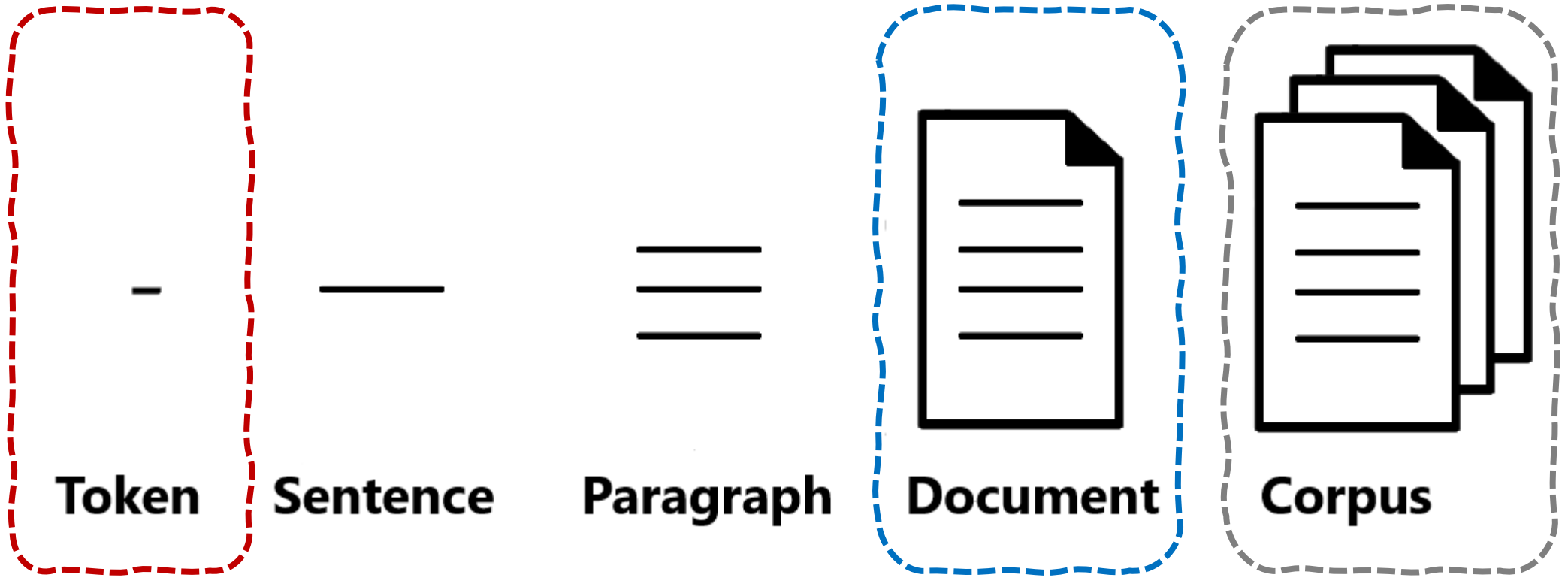
D1	“The delivery was late and my package was delayed.”
D2	“The package arrived late. I requested a refund.”
B1	“My subscription payment failed this month.”
P1	“The product arrived damaged and broken.”

D: Delivery

B: Billing

P: Product

Three Building Blocks



Three Building Blocks

- **Corpus**
  - The complete collection of documents to analyze.
- **Document**
  - One unit of text that becomes one observation.
- **Token**
  - A unit produced by a tokenizer: a word, subword, or other text fragment.

Tokenization

- Original ticket

> “The package arrived late. I requested a refund.”

- Illustrative word-level tokens

the	package	arrived	late	i	requested	a	refund

Preprocessing

- **Lowercasing**
- **Removing some punctuation**
- **Stop words filtering**
  - Removes frequently occurring words that may provide limited information for distinguishing documents.
  - Reduces unnecessary features and computational complexity.
  - **Stop word examples: “**the,” “a,” “an,” “is,” “and,” “of”
  - **Sentence examples:**
    - Originally, “the product is delivered to the customer.”
    - After filtering, “product delivered customer”

Preprocessing

- **Stemming or Lemmatization**
  - Converts different forms of a word into a common representation, reducing vocabulary size and grouping related words.

Feature	Stemming	Lemmatization
Approach	Rule-based (heuristic)	Dictionary-based (linguistic)
Output Type	Root word, but often not a valid word	Proper base form found in dictionaries
Grammar Awareness	No grammatical awareness	Considers parts of speech (POS)
Processing Time	Faster (simple heuristic rules)	Slower (requires dictionary lookup)
Accuracy	Less accurate, sometimes produces meaningless words	More accurate, always returns correct words
Example (flies)	"fli" (incorrect)	"fly" (correct)
Example (better)	"better" (unchanged)	"good" (correct lemma)
Best Used For	Quick text normalization, search indexing	High-accuracy NLP tasks like chatbot processing and sentiment analysis

Preprocessing

- **Preprocessing is a choice, not a universal rule**
  - **Common options**
    - Lowercasing
    - Removing some punctuation
    - Stopword filtering
    - Stemming or lemmatization
  - **Beware of losing meaning**
    - “I am satisfied.”
    - “I am NOT satisfied.”

Smaller than a token?

- **Characters vs. Sub-characters in Some Languages**

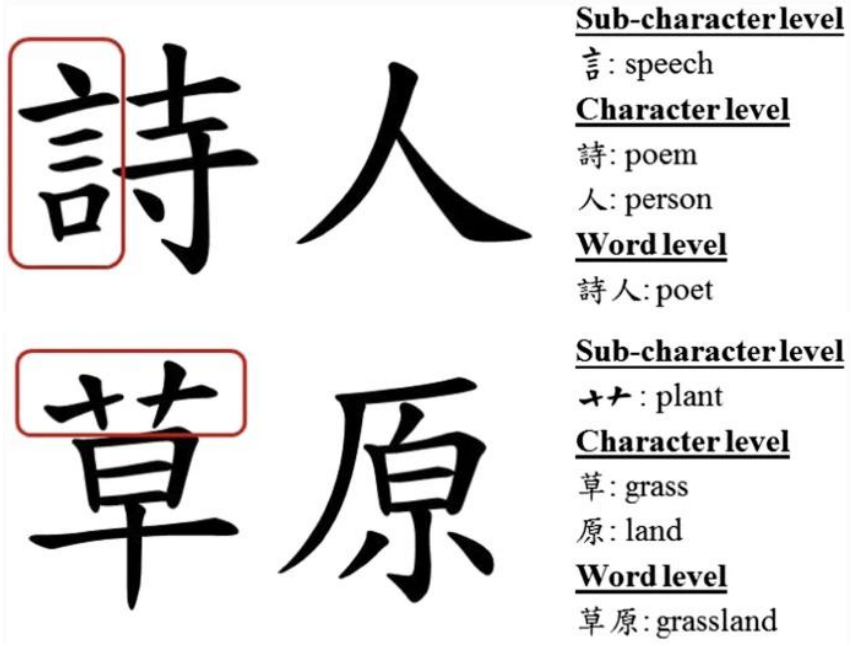
Hangeul or **Korean alphabet**
(*Phonogram*)



Figure 1: Sub-character Example of Korean characters.

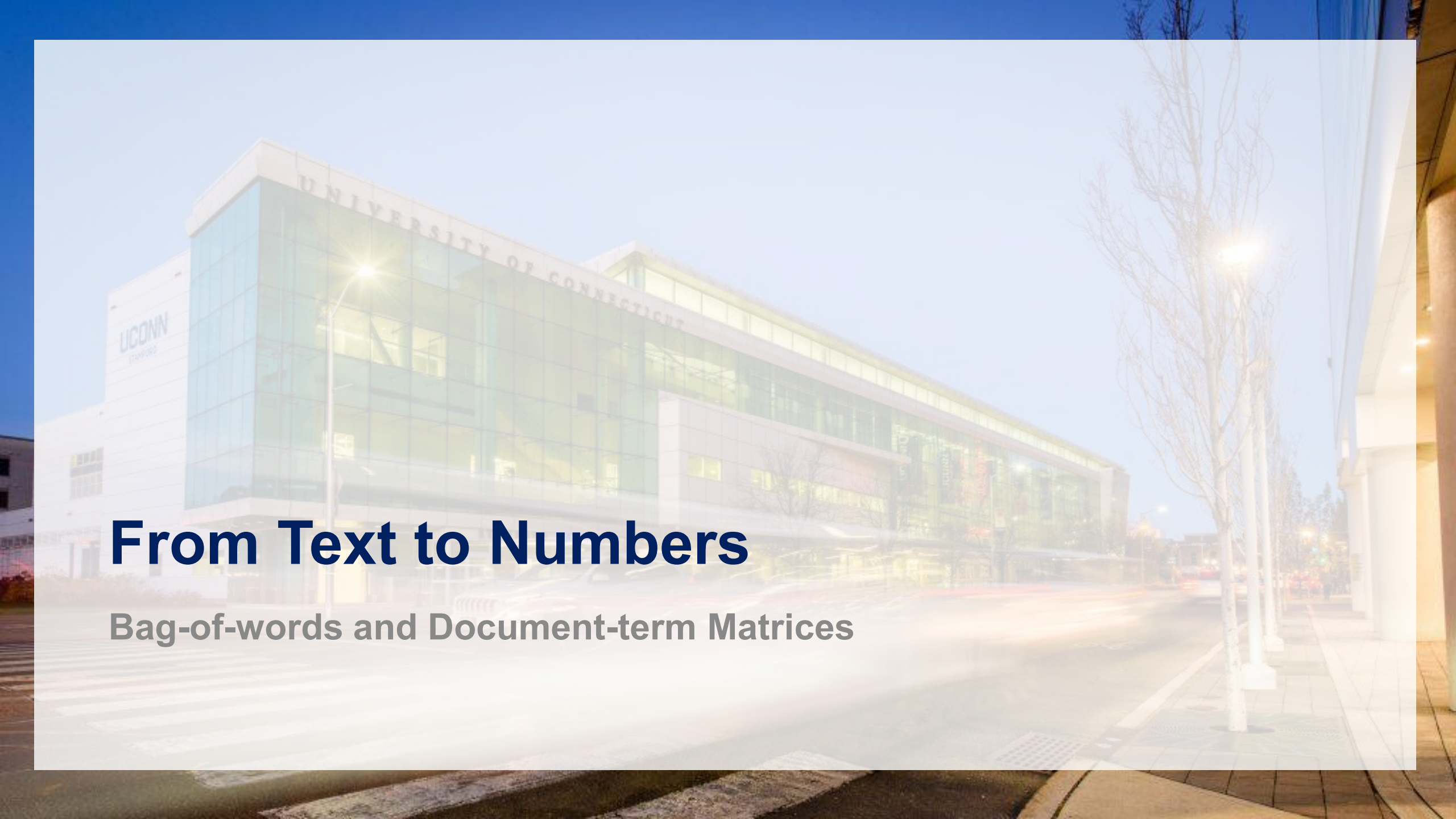
J-eo-ng (3) **Ch-i** (2)

Chinese Character
(*Logogram*)



Sources

- TaeYoung Kang and **Jaeung Sim** (2021) "Partisan Responses to Fact-Checking in Online News Platforms: Evidence from a Political Rumor about the North Korean Leader," *Proceedings of the Fifteenth International AAAI Conference on Web and Social Media (ICWSM)*, pp. 266-27.
- Catherine McBride, Ying Wang, and Leo Man-Lit Cheang (2018) "Dyslexia in Chinese," *Current Developmental Disorders Reports* 5(4), pp. 217-225.



From Text to Numbers

Bag-of-words and Document-term Matrices

Bag-of-Words (BoW)

I love this movie! It's sweet, but with satirical humor. The dialogue is great and the adventure scenes are fun... It manages to be whimsical and romantic while laughing at the conventions of the fairy tale genre. I would recommend it to just about anyone. I've seen it several times, and I'm always happy to see it again whenever I have a friend who hasn't seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
…	…

Bag-of-Words (BoW)

- Two steps transform a document into a vector.
  - **Step 1. Create vocabulary**
    - One column for each selected item.
  - **Step 2. Count occurrences**
    - One row (vector) for each document.
- Example:

Document D2: “The package arrived late. I requested a refund.”

Vocabulary: [package, late, refund, subscription, payment]

D2 count vector: [1, 1, 1, 0, 0]

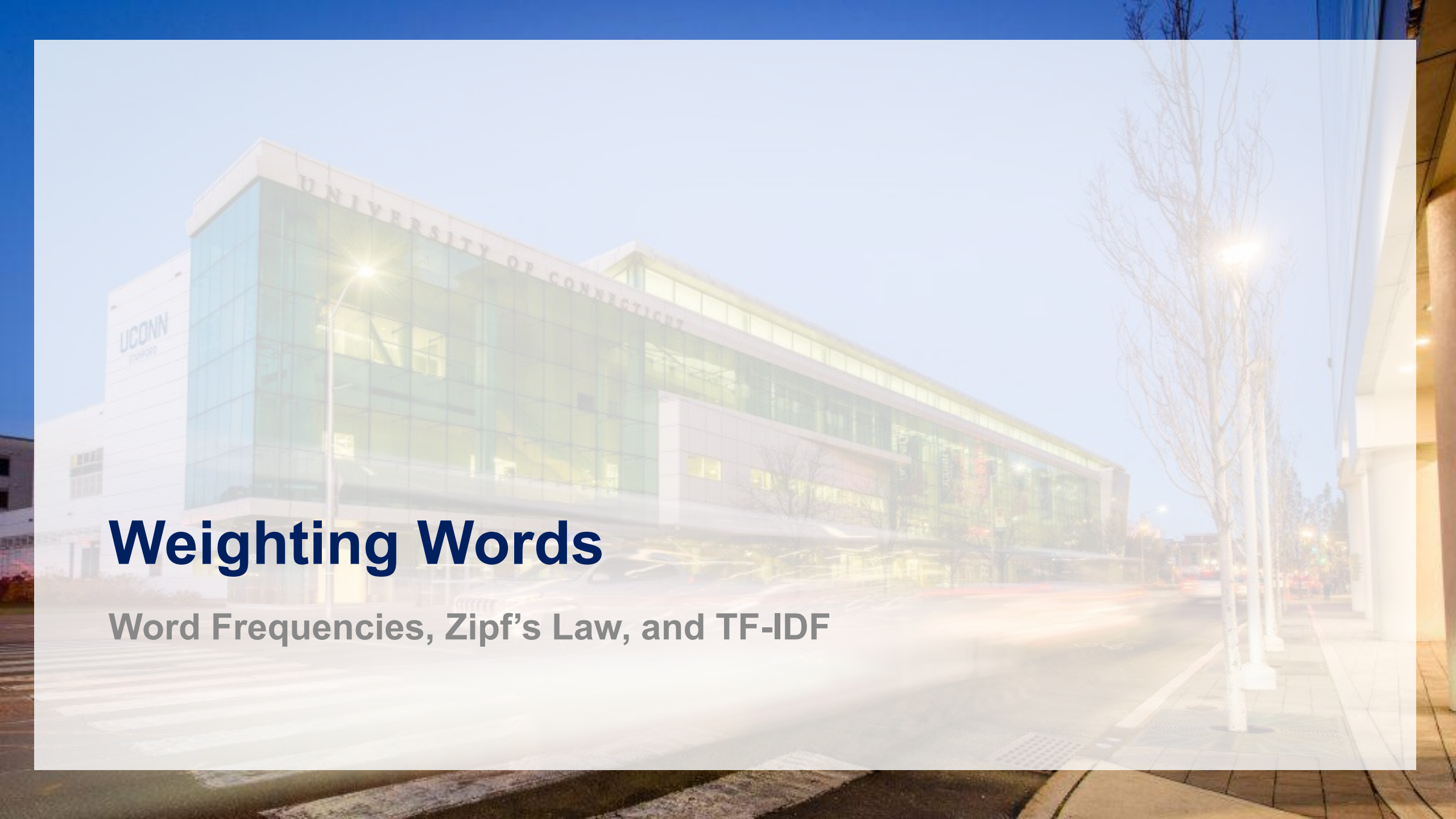
Document-term Matrix

- **Documents as Observations**
  - Rows = documents; columns = terms; each cell = number of occurrences.

Ticket	package	late	refund	subscription	payment	product	damaged
D1	1	1	0	0	0	0	0
D2	1	1	1	0	0	0	0
B1	0	0	0	1	1	0	0
B2	0	0	0	1	0	0	0
P1	0	0	0	0	0	1	1
P2	0	0	0	0	0	1	0

Limitations

- **Bag-of-words ignores word order.**
  - Example:
    - “The customer does not like the service.”
    - “The service does not like the customer.”
    - The two sentences have the identical unigram word counts, although the roles of “customer” and “service” have changed.
- **Text data becomes high-dimensional.**
  - **Many possible terms**
    - A vocabulary of 10,000 words implies 10,000 columns—even when one ticket is only a sentence long.
  - **Many zeros**
    - Each ticket contains only a tiny fraction of the vocabulary. Most matrix cells are zero (sparsity).

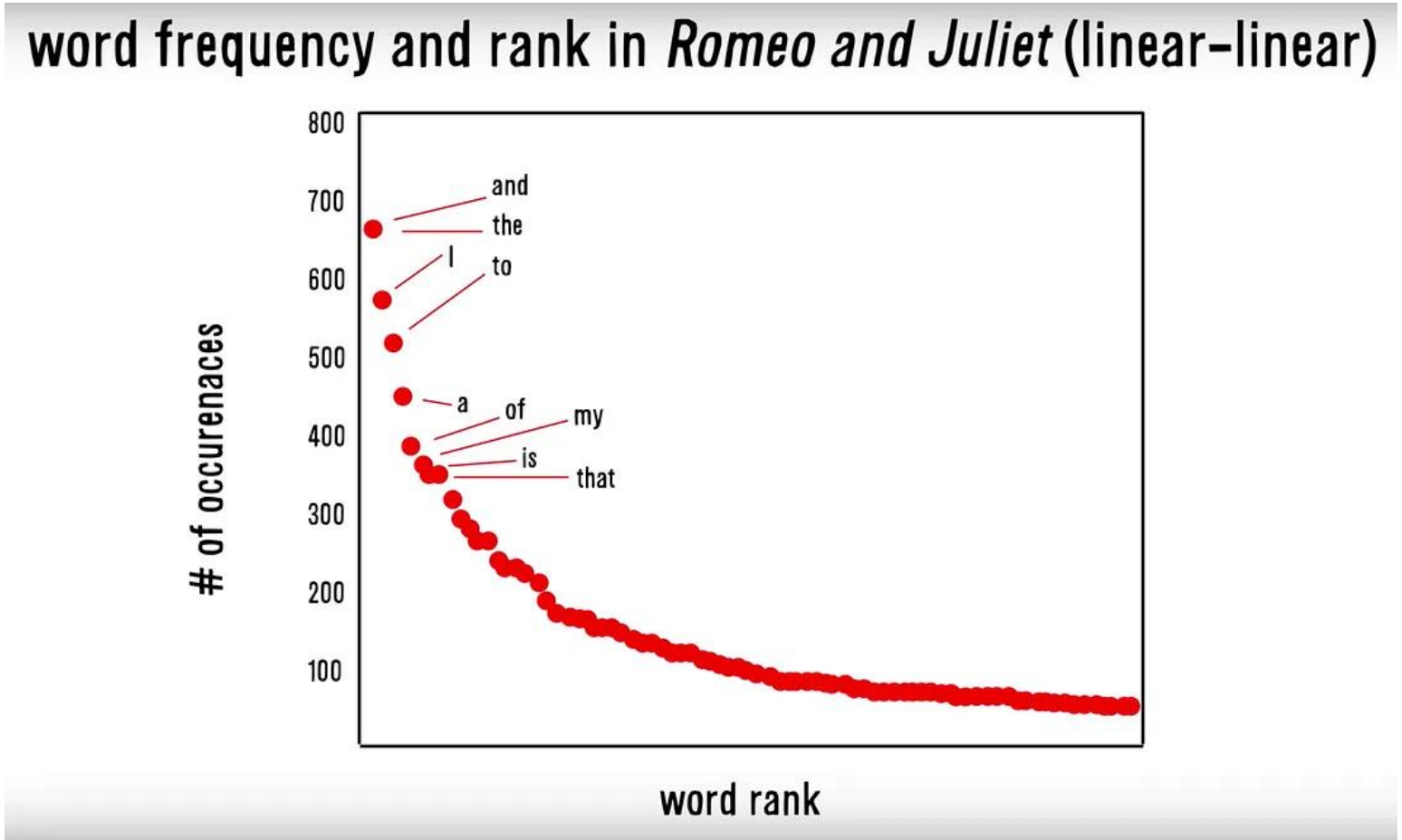


Weighting Words

Word Frequencies, Zipf's Law, and TF-IDF

Zipf's Law

- Natural Language Has a Long Tail



- A few terms occur often.
- Many terms occur rarely.
- Raw term counts are highly uneven.

TF-IDF

- **TF-IDF = Term Frequency (TF) × Inverse Document Frequency (IDF)**
  - **Term frequency (TF)**
    - How many times does the term occur in this document?
  - **Inverse document frequency (IDF)**
    - How uncommon is this term across documents in the corpus?
  - Count terms, then adjust for distinctiveness.
    - A frequent word in one ticket can be less distinctive if it appears in nearly every ticket.

TF-IDF

- A Simple IDF Calculation
  - **IDF(term) = ln(total number of documents / documents containing the term)**

Term	Documents with term	IDF = ln(12 / df)	Intuition
package	4 of 12	1.10	More common
subscription	3 of 12	1.39	Moderately distinctive
refund	1 of 12	2.48	**Most distinctive here**

TF-IDF

- **Before and After: Counts vs. TF-IDF Weights**

Raw word counts

	package	refund	subscription	product
D1	1	0	0	0
D2	1	1	0	0
B1	0	0	1	0
P1	0	0	0	1

TF-IDF weights

	package	refund	subscription	product
D1	1.10	0.0	0.0	0.0
D2	1.10	2.48	0.0	0.0
B1	0.0	0.0	1.39	0.0
P1	0.0	0.0	0.0	1.39

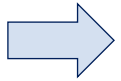
TF-IDF

- What do we do with TF-IDF weights?
  - TF-IDF replaces raw counts with numerical features that emphasize relatively distinctive terms.
  - TF-IDF is a representation, not a model. After weighting, the next question is whether we need all of these term columns—or whether SVD can summarize them using fewer dimensions.

Illustrative TF-IDF matrix

Document	delivery	refund	subscription
Review 1	0.3	**0.8**	0.0
Review 2	**0.6**	0.0	0.0
Review 3	0.0	0.0	**0.9**

Each row is now a numerical vector for one document.
We can compare, group, or model documents using these features.



Once text becomes numbers, we can…

Document similarity
Compare TF-IDF vectors to find documents with similar informative terms.

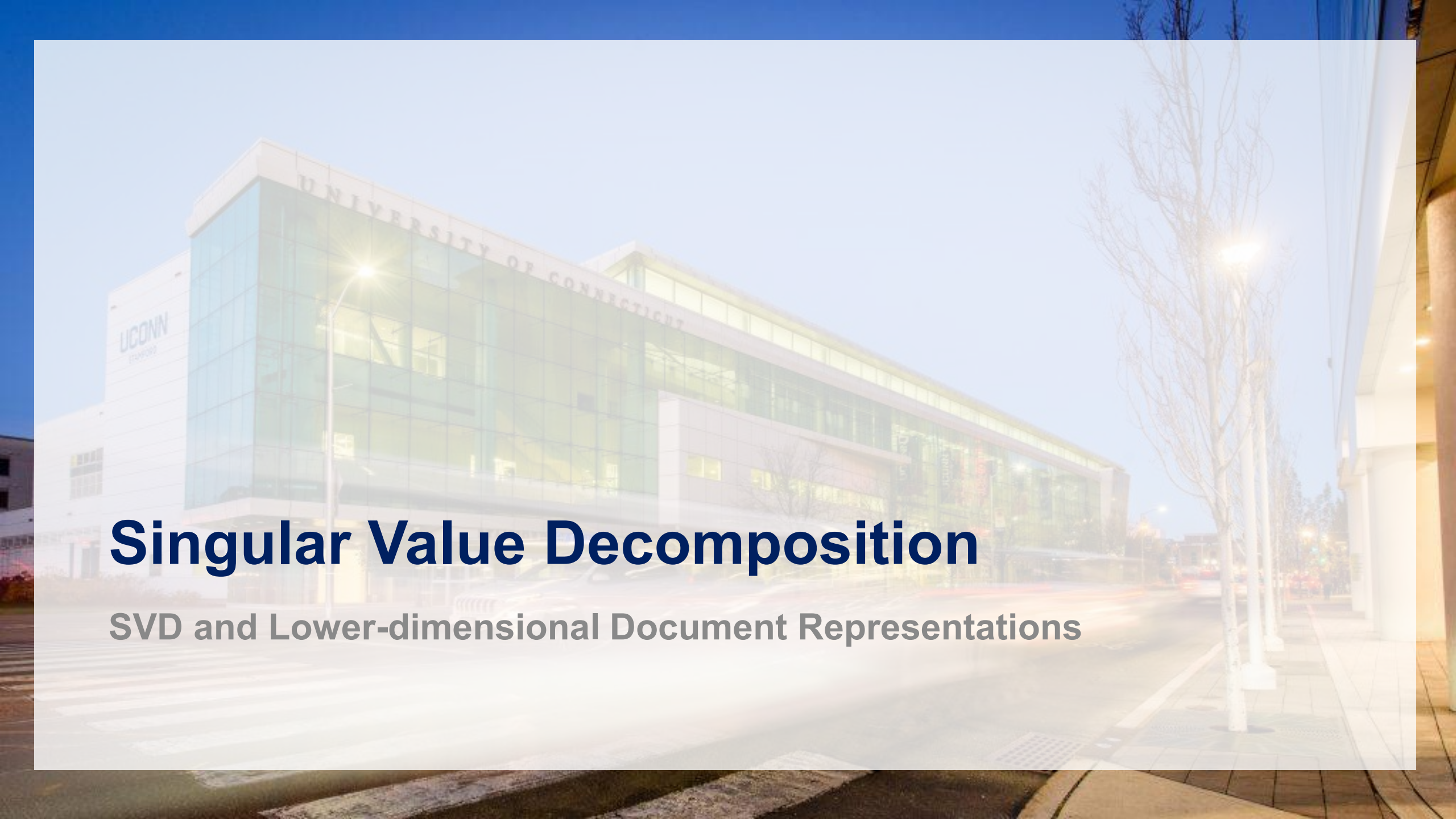
Clustering
Group documents with similar vocabulary patterns.

Prediction
Use TF-IDF columns as inputs to classification or regression models.

Dimensionality reduction
Apply SVD to compress thousands of weighted terms into fewer features.

TF-IDF

- Captures:
  - Word occurrence patterns
  - Term distinctiveness
  - A clear numerical baseline
- Can miss:
  - Word order and nuanced context
  - Synonyms with different spellings
  - Negation and implicit meaning
- IMPORTANT: High TF-IDF does NOT automatically mean important, correct, or business-relevant.



Singular Value Decomposition

SVD and Lower-dimensional Document Representations

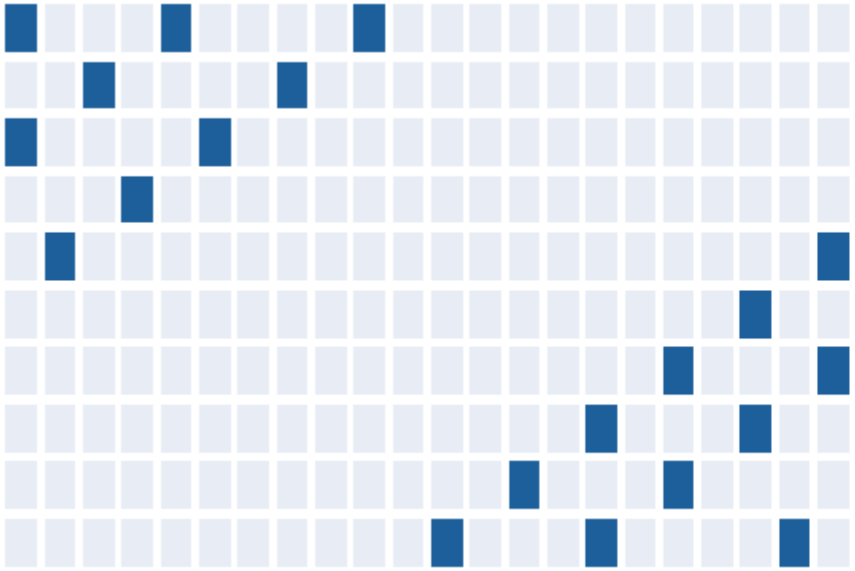
Sparse, High-Dimensional Vector Spaces

- **# unique terms >> # documents in general**
  - After the frequency counts are obtained, you see that both terms and documents can be represented in vector spaces.
  - However, in both cases, even after stemming and other filtering steps have been applied, you usually still face a very high-dimensional data set.
  - In addition, the matrices of frequency counts are very sparse because many words appear in only one or two documents. Typically, 90% or more of the cells in the matrices are 0.
  - Also, the frequency counts are highly skewed, as shown by Zipf's law; that is, a small number of words occur many times.

Singular Value Decomposition

- Summarizing a high-dimensional matrix

Documents represented with many terms



Many documents × many terms

Documents represented With a few dimensions

+0.00	+0.99
+0.78	+0.52
+0.97	-0.35
+0.43	-0.95
-0.44	-0.83
-0.98	-0.08
-0.77	+0.73
+0.02	+0.99
+0.79	+0.50
+0.97	-0.37

Many documents (same) × 2 dimensions

Singular Value Decomposition

- A minimal mathematical view of (truncated) SVD

$$X \approx U_k \Sigma_k V_k^T$$

X

Original document-term matrix (documents × terms)

Keep k dimensions

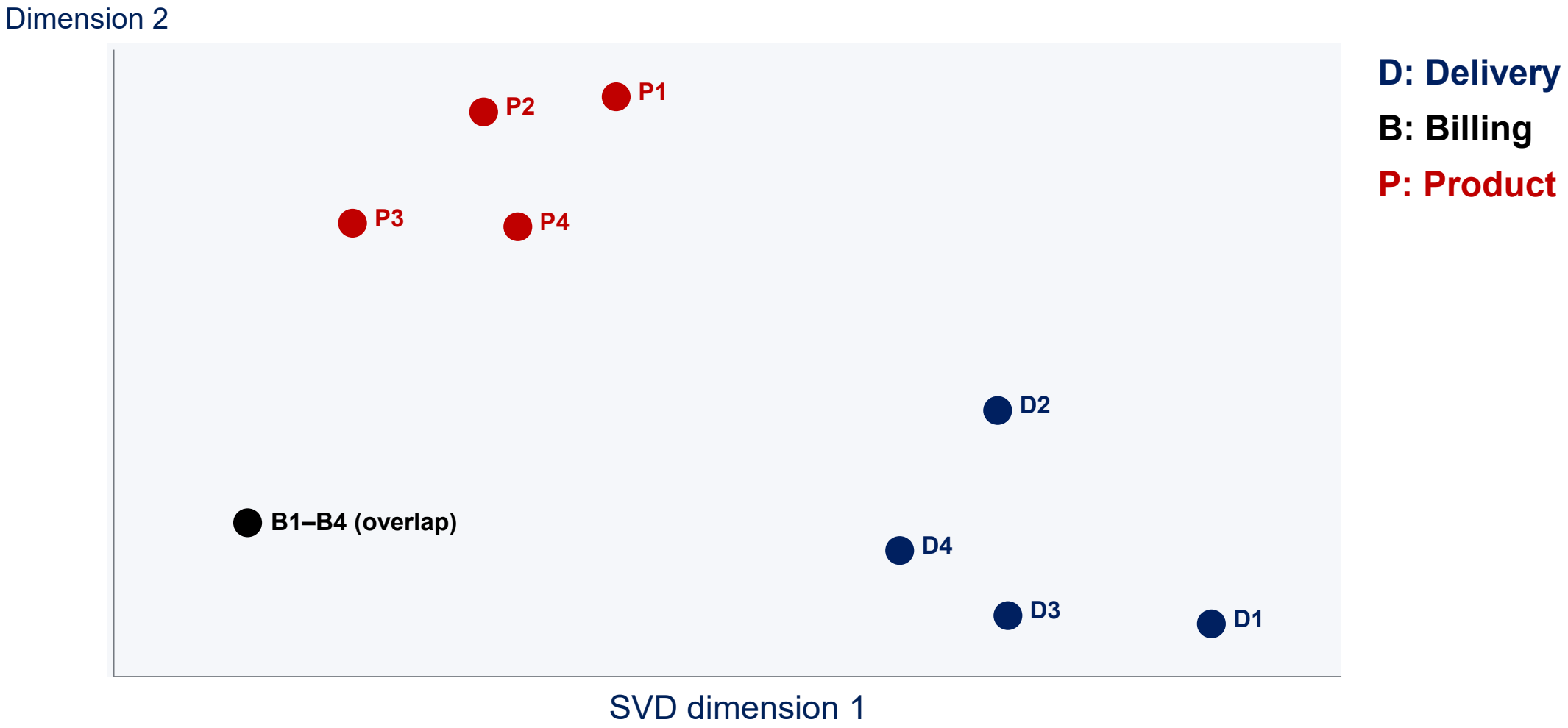
Preserve the strongest selected patterns, not every detail.

Reduced features

Each document gets k numerical coordinates.

Singular Value Decomposition

- Document map after truncated SVD (2 Dimensions)

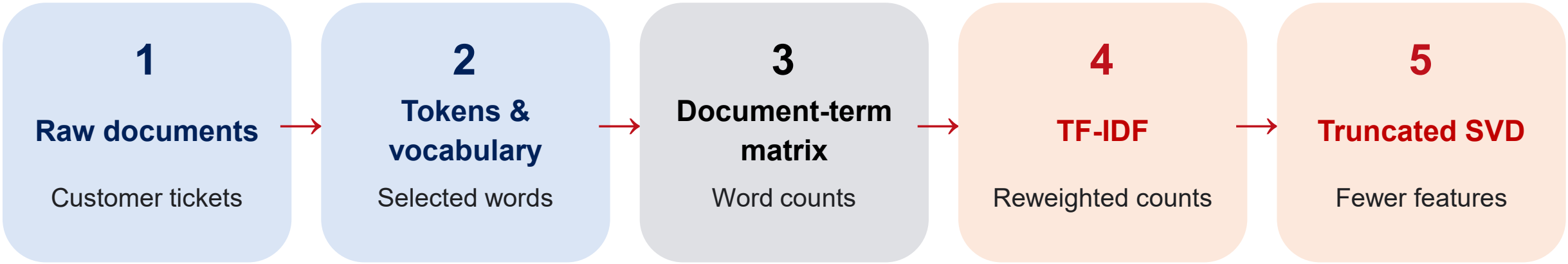


Singular Value Decomposition

- **How to interpret an SVD map**
  - Reasonable interpretation
    - Nearby documents may share patterns in their numerical term representations.
  - Avoid overclaiming
    - A two-dimensional map may lose information.
    - SVD dimensions are not guaranteed to be clean topics.
  - Keeping more dimensions usually preserves more of the original matrix's structure, but it reduces compression.

One Complete Text-Mining Workflow

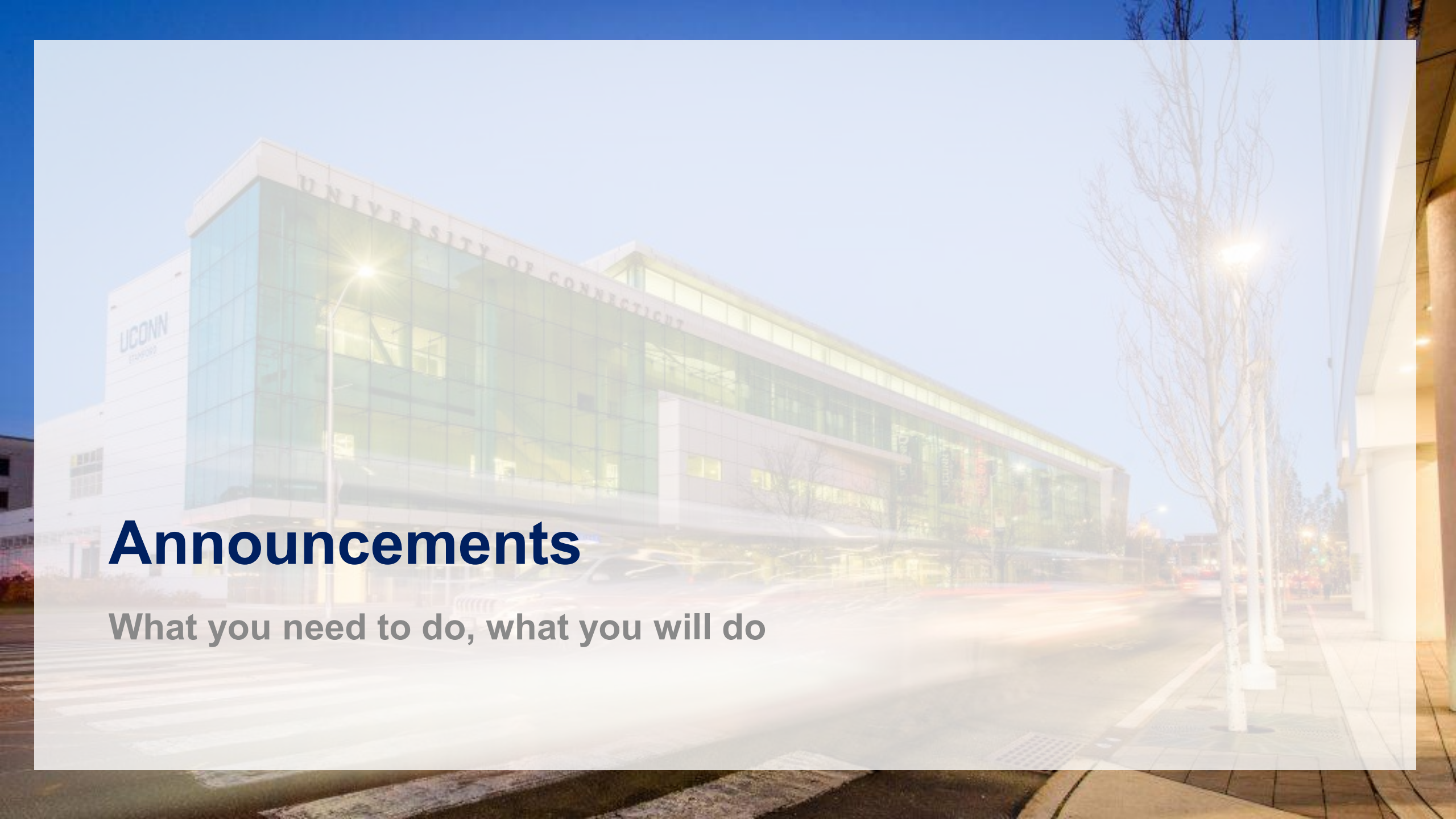
- **Objective:** Transform text into useful numerical features for analysis and prediction.



Live Quiz

- **Join the quiz!**
  - https://pollev.com/jaeungsim088





Announcements

What you need to do, what you will do

Announcements

- **Oct 27 (Tue), 2026 (Next week)**
  - **Feedback on the Take-home Midterm Exam**
  - Python Hands-on for Quantifying Texts
  - Python Hands-on for Supervised and Unsupervised Feature Generation
  - **Hands-on Homework #2 will be released.**
  - **Team Project #2 will be released.**
- **Nov 3 (Tue), 2026: Deep Learning and Text Mining**
  - **Nov 8 (Sun), 2026: Homework #2 Deadline**
- **Nov 10 (Tue), 2026: Recent Challenges in Machine Learning**
- **Nov 17 (Tue), 2026 (Virtual): Term Project Presentation #2**
- **Nov 24 (Tue), 2026 (No Class): *Thanksgiving Recess***
- **Dec 1 (Tue), 2026 (Asynchronous): Take-home final exam will be released.**



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